

StringConcatTrap.java

for (int n=10000; n <= 80000; n*=2)

n=10,000

10,000 < 80,000 ✓ (1)

n ≠ n*2

= 10000 * 2

n=20,000

20,000 < 80,000 ✓ (2)

n=40,000

40,000 < 80,000 ✓ (3)

n=80,000

80,000 < 80,000 ✗ ✓ (4)

n=160,000

160,000 < 80,000 ✗

not executed

loop run for (4) times

Suppose

n=10,000

initially

x = ""

iteration

i=0 → S = "x"

i=1 → S = "xx"

i=2 → S = "xxx"

i=9999 → S = "contain 10,000 x's"

S.length = 10,000



ADSUL'S TECHNICAL CAMPUS



String are immutable in java

so when

`s += 'X';`

new string is created,

it not modify existing string (slow)

String Builder

`n = 10,000`

so `append('x')`

$\vdots$

it create concatenation

so complexity $O(n)$

`s.length = 10000`

`sb.length = 10,000` (faster)

n	s length	sb.length
10,000	10,000	10,000
20,000	~	~
40,000	~	~
80,000	~	~

in string n get double so complexity $O(n^2)$

in String Builder complexity $2x = O(n)$

const ignore:

